MTH2010 Algebra

Graded Formative Sheet 1: Solutions

1. (a) (i) $\mathbf{Z}_8$ has a unique subgroup of cardinality 4 which is

$$H = \{[0], [2], [4], [6]\}$$

generated by its elements of order 4: [2] and [6].
This subgroup has 2 left cosets:
$[2] + H = [4] + H = [6] + H = [0] + H = H$
and
$[1] + H = [3] + H = [5] + H = [7] + H = \{[1], [3], [5], [7]\}$.

(ii) $\mathbf{Z}_{12}$ has a unique subgroup of cardinality 6 which is

$$H = \{[0], [2], [4], [6], [8], [10]\}$$

generated by the element of order 6: [2].
This subgroup has 2 left cosets:
$[0] + H = [2] + H = [4] + H = [6] + H = [8] + H = [10] + H = H$
and
$[1] + H = [3] + H = [5] + H = [7] + H = [11] + H = \{[1], [3], [5], [7], [9], [11]\}$.

(b) • $H \stackrel{\text{def}}{=} \langle sr^2 \rangle = \{e, sr^2\}$ has 4 left cosets: $H = eH = sr^2H$, $rH = srH = \{r, sr\}$, $r^2H = sH = \{s, r^2\}$, and $r^3H = sr^3H = \{r^3, sr^3\}$.

• $G = \langle r \rangle = \{e, r, r^2, r^3\}$ has 2 left cosets: $G = eG = rG = r^2G = r^3G$ and $sG = srG = sr^2G = sr^3G = \{s, sr, sr^2, sr^3\}$.

2. (a) Let

$$D_{2n} = \{1, r, \ldots, r^{n-1}, s, sr, \ldots, sr^{n-1}\}$$

be the dihedral group with $2n$ elements with

$$r^n = s^2 = e \qquad \text{and} \qquad sr^j = r^{n-j}s \qquad 0 \leq j \leq n-1,$$

and let

$$D_{2k} = \{1, r', \ldots, (r')^{k-1}, s', s'r', \ldots, s'(r')^{k-1}\}$$

be the dihedral group with $2k$ elements with

$$(r')^k = (s')^2 = e \qquad \text{and} \qquad s'(r')^j = (r')^{k-j}s' \qquad 0 \leq j \leq k-1.$$

Define a map

$$\phi : D_{2n} \to D_{2k}$$

by

$$\phi(r) = r'$$

and

$$\phi(s) = s'.$$

Thus ϕ is defined, for $0 \leq j \leq 1$ and $0 \leq i \leq n-1$, by:

$$\phi(s^j r^i) = (s')^j (r')^i.$$

This map is surjective: if $(s')^i(r')^j \in D_{2k}$ then

$$(s')^i(r')^j = \phi(s^i r^j),$$

and it is a homomorphism:

$$\phi(sr^i sr^l) = \phi(s^2 r^{l-i}) = \phi(r^{l-i}) = (r')^{l-i} = (s')^2(r')^{l-i} = s'(r')^i s'(r')^l = \phi(sr^i)\phi(sr^j),$$

and likewise one shows easily that

$$\phi(sr^i r^l) = \phi(sr^i)\phi(r^l),$$

$$\phi(r^i sr^l) = \phi(r^i)\phi(sr^l),$$

and

$$\phi(r^i r^l) = \phi(r^i)\phi(r^l).$$

We have $\phi(r^k) = (r')^k = e$. Hence $r^k \in \operatorname{Ker}(\phi)$ and $H = \langle r^k \rangle \subset \operatorname{Ker}(\phi)$. Since $2n = |D_{2n}| = |\operatorname{Ker}(\phi)||D_{2k}| = |\operatorname{Ker}(\phi)|2k$ by the first isomorphism theorem which states $D_{2n}/\operatorname{Ker}(\phi)$ is isomorphic to D_{2k}, we see that $|\operatorname{Ker}(\phi)| = \frac{n}{k}$. Further $\operatorname{ord}(r^k) = |H| = \frac{n}{k}$. Hence the equality $H = \operatorname{Ker}(\phi)$ must hold as these groups have the same cardinality and $H \subset \operatorname{Ker}(\phi)$.

(b) The first isomorphism theorem (FIT) implies (indeed states in this case) that $D_{2n}/\operatorname{Ker}(\phi) = D_{2n}/H$ is isomorphic to $\operatorname{Im}(\phi) = D_{2k}$ since ϕ is a surjective homomorphism.

3. (a) I present **two** different solutions to this problem.

First solution: Suppose, for a contradiction, that there is a surjective homomorphism $f : A_4 \to \mathbb{Z}/2/Z$. Then $\operatorname{Ker} f$ contains every 3-cycle. Indeed, for every three-cycle $\sigma \in A_4$, we have $\sigma^3 = e$, whence $0 = f(e) = f(\sigma^3) = 3f(\sigma)$, so the order of $f(\sigma)$ is a factor of 3. But, by Lagrange's Theorem, the order of $f(\sigma)$ is a factor of 2. So the order of $f(\sigma)$ is 1. Thus, $f(\sigma) = e$.
But there are eight 3-cycles in A_4. So $|\operatorname{Ker}(f)| \geq 8$. But, as $|ker(f)|$ divides $12 = |A_4|$, it follows that $|\operatorname{Ker}(f)| = 12$. By the first isomorphism theorem, $12 = |A_4| = |\operatorname{Ker}(f)| \cdot |\operatorname{Im}(f)| = 12 \times 2 = 24$, as f is surjective. This is a contradiction.
A surjective homomorphism from A_4 to $\mathbb{Z}/2\mathbb{Z}$ must have for kernel a normal subgroup of cardinality 6. But A_4 has no (normal) subgroups of cardinality 6 so there is no such homomorphism.

Second solution: Suppose, for a contradiction, that there is a surjective homomorphism $f : A_4 \to \mathbb{Z}/2\mathbb{Z}$. Then, by the first isomorphism theorem, $12 = |A_4| = |\operatorname{Ker}(f)| \cdot |\operatorname{Im}(f)| = 2|\operatorname{Ker}(f)|$, whence $|ker(f)| = 6$. But $\operatorname{Ker}(f)$ is a normal subgroup of A_4. To arrive at a contradiction, it suffices to show that A_4 has no normal subgroup of cardinality 6.
Indeed A_4 has 3 elements of order 2 (all product of 2 disjoint transpositions) and 8 elements of order 3 (all 3 cycles). Suppose H is a subgroup of cardinality $6 = 2 \times 3$ of A_4. Then H must contain an element of order 2 (a product of 2 transposition) and a 3 cycle (by Cauchy's theorem for the primes 2 and 3 which divide $|A_4| = 12$). In particular H must contain all product of 2 transpositions as these are all conjugate in A_4 (check this fact, seen in Lecture). Hence H contains the subgroup $\{e, (12)(34), (13)(24), (14)(23)\}$ of cardinality 4 which is not possible since 4 doesn't divide $|H| = 6$.

(b) A surjective homomorphism $\psi : A_4 \to \mathbb{Z}/3\mathbb{Z}$ must have a kernel of cardinality 4. The subgroup $H = \{e, (12)(34), (13)(24), (14)(23)\}$ is a normal subgroup of A_4 of order 4. We can construct ψ as follows

$$\psi(\sigma) = \{\bar{0}\} \text{ for } \sigma \in H,$$

$$\psi((123)) = \bar{1}, \quad \psi((132)) = \bar{2}, \quad \psi((124)) = \bar{2}, \quad \psi((142)) = \bar{1}, \quad \psi((134)) = \bar{1},$$
$$\psi((143)) = \bar{2}, \quad \psi((234)) = \bar{2}, \quad \psi((243)) = \bar{1}.$$

Check that ψ is a homomorphism, it is surjective by construction.
Alternatively, work out the elements of A_4/H, construct an isomorphism $\phi : A_4/H \to \mathbb{Z}/3\mathbb{Z}$, and consider the composite $\psi : A_4 \to A_4/H \xrightarrow{\phi} \mathbb{Z}/3\mathbb{Z}$ where the first map $A_4 \to A_4/H$ is the surjective homomorphism mapping $\sigma \in A_4$ to σH.

4. (a) First, we show that $H \cap K$ is a subgroup. Note $e_G \in H$ and $e_G \in K$ so $e_G \in H \cap K$. Thus $H \cap K$ is non empty. If $x \in H \cap K$ then $x^{-1} \in H$ and $x^{-1} \in K$, since H and K are subgroups of G, hence $x^{-1} \in H \cap K$. Similarly, if $x, y \in H \cap K$ then $xy \in H$ and $xy \in K$ since H and K are subgroups of G hence $xy \in H \cap K$. **(6 marks)**
Assume now that both H and K are subgroups of G. Then, using the previous part of the question, $H \cap K$ is a subgroup of G Let $x \in H \cap K$ and let $y \in G$. Then $yxy^{-1} \in H$ and $yxy^{-1} \in K$ since both H and K are normal. Thus $yxy^{-1} \in H \cap K$ And $H \cap K$ is a normal subgroup of G. **(7 marks)**

(b) Let $h \in H$ and $g \in N \cap H$. Now $h \in G$ (as $H \subseteq G$), so, as N is normal in G, we have $hgh^{-1} \in N$. Now $g \in N \cap H$ so $g \in H$. Thus, $hgh^{-1} \in H$, since $g, h \in H$ and H is a subgroup. Thus $hgh^{-1} \in N \cap H$, and $N \cap H$ is normal in H. **(7 marks)**

5. (a) First note that $(e_A, e_B) \in \tilde{A}$ hence $\tilde{A}$ is nonempty.
Let $a, a' \in A$ and $b \in B$. Then

$$(a, e_B)(a', e_B) = (aa', e_B) \in \tilde{A}$$

and

$$(a, e_B)^{-1} = (a^{-1}, e_B) \in \tilde{A},$$

which shows $\tilde{A}$ is a subgroup.
Further

$$(a, b)(a', e_B)(a, b)^{-1} = (a, b)(a', e_B)(a^{-1}, b^{-1}) = (aa'a^{-1}, be_Bb^{-1}) = (aa'a^{-1}, e_B) \in \tilde{A}$$

since $aa'a^{-1} \in A$, hence $\tilde{A} = \{(a, e_B) : a \in A\}$ is a normal subgroup of $A \times B$.
Consider the map $\phi : A \times B \to B$ defined by

$$\phi((a, b)) = b$$

which is surjective. The map ϕ is a homomorphism:

$$\phi((a, b)(a', b')) = \phi(aa', bb') = bb' = \phi((a, b))\phi((a, b')).$$

The kernel of ϕ is precisely

$$\mathrm{Ker}(\phi) = \{(a, b) : \phi(a, b) = b = e_B\} = \tilde{A} = \{(a, e_B) : a \in A\}.$$

The first isomorphism theorem implies then that the factor group $(A \times B)/\tilde{A}$ is isomorphic to B.

(b) Let $a_1, a_2 \in A$ then

$$a_1B.a_2B = a_1a_2B = a_2a_1B = a_2B.a_1B$$

where we used the fact that A is abelian for the second equality. Hence A/B is abelian.

6. Let $b = g \cdot a$, $g \in G$. We first show $g\,\mathrm{Stab}(a)g^{-1} \subset \mathrm{Stab}(b)$. Let $h \in \mathrm{Stab}(a)$, then

$$(ghg^{-1})b = (ghg^{-1})(g \cdot a) = (gh)(g^{-1}g \cdot a) = (gh) \cdot a = g \cdot (h \cdot a) = g \cdot a = b$$

[we used the fact $h \cdot a = a$].
Next we show $\mathrm{Stab}(b) \subset g\,\mathrm{Stab}(a)g^{-1}$. Let $h \in \mathrm{Stab}(b)$, i.e. $h \cdot b = b$. Note that $b = g \cdot a$ implies $g^{-1} \cdot b = g^{-1}(g \cdot a) = a$. Then $h' \overset{\text{def}}{=} g^{-1}hg \in \mathrm{Stab}(a)$. Indeed

$$h' \cdot a = (g^{-1}hg) \cdot a = (g^{-1}h) \cdot (g \cdot a) = (g^{-1}h) \cdot b = g^{-1} \cdot (h \cdot b) = g^{-1} \cdot b = a.$$

Thus $h = gh'g^{-1} \in g\,\mathrm{Stab}(a)g^{-1}$.
If G acts transitively on A then there is only one orbit say $A = \mathrm{orb}(a)$ and every element b of A equals $g \cdot a$ for some $g \in G$. In this case all stabilisers of elements of A are conjugate to $\mathrm{Stab}(a)$ since for $b \in A$, if $b = g \cdot a$, then $\mathrm{Stab}(b) = g\,\mathrm{Stab}(a)g^{-1}$ as seen above.
Thus the kernel of the action, which consists of elements of G which belong to every stabiliser of all elements of A; i.e. $\bigcap_{b \in A} \mathrm{Stab}(b)$, is equal

$$\bigcap_{b \in A} \mathrm{Stab}(b) = \bigcap_{g \in G} g\,\mathrm{Stab}(a)g^{-1}.$$

7. The left cosets of H in D_8 are:

$$\begin{aligned} H = sH &= \{s, 1\} \\ rH = sr^3H &= \{sr^3, r\} \\ r^2H = sr^2H &= \{r^2, sr^2\} \\ r^3H = srH &= \{r^3, sr\} \end{aligned}$$

which we label as $1, 2, 3, 4$, respectively.
The action of G on these left cosets by (left) multiplication gives rise to a representation [a homomorphism] $\pi_H : D_8 \to S_4$ which we compute. To compute $\pi_H(r)$ we multiply the left cosets of H by r on the left to obtain:

$$\begin{aligned} &rH \\ r(rH) &= r^2H \\ r(r^2H) &= r^3H \\ r(r^3H) &= H \end{aligned}$$

thus with the labelling above $\pi_H(r)$ maps the coset labelled 1 to the coset labelled 2, and 2 to 3, 3 to 4 and 4 to 1 hence this is the 4-cycle (1234) of S_4.
In a similar way we can compute (check this!)

$$\begin{aligned} &\pi_H(1) = e_{S_4}, \quad \pi_H(r) = (1234), \quad \pi_H(r^2) = (13)(24), \quad \pi_H(r^3) = (1432) \\ &\pi_H(s) = (24), \quad \pi_H(sr) = (14)(23), \quad \pi_H(sr^2) = (13), \quad \pi_H(sr^3) = (12)(34). \end{aligned}$$

In particular this representation is faithful as π_H is injective. The first isomorphism theorem implies that $D_8/\mathrm{Ker}(\pi_H)$, which is isomorphic to D_8 since $\mathrm{Ker}(\pi_H) = \{1\}$, is isomorphic to $\pi_H(D_8)$. Thus $\pi_H(D_8)$ is a subgroup of S_4 isomorphic to D_8.